

TECHNICAL & ECONOMIC WHITEPAPER

The Inaya Protocol

A decentralized sovereign custody network for high-value enterprise and personal data assets.

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SECTION 01

Abstract

Contemporary high-value enterprise data storage carries a structural single-point-of-failure risk: centralized cloud platforms hold complete, whole files, meaning a single breach of infrastructure or credentials exposes the entire asset.

The Inaya Protocol eliminates this by never letting a complete, readable copy of a file exist anywhere outside the owner's own device. Every file is encrypted locally, split into two independent shards, and pinned to decentralized storage — with only a tamper-evident ownership record, never the file itself, ever touching the blockchain.

This document covers the cryptographic mechanics, the on-chain custody model, the token economics, and — new since the protocol's original whitepaper — the application layer now built on top of it: a Business SaaS workspace, a decentralized security/threat-intelligence layer, an educational platform, an investor data room, two desktop apps, and three purpose-built AI assistants. All of it runs on BNB Chain Testnet today; nothing in this document describes a mainnet-live system yet.

SHARD VECTORS PER ASSET	ENCRYPTION	\$INAYA TOTAL SUPPLY
2	256-bit AES-GCM	30,000,000 (fixed)

SECTION 02

The Problem With Centralized Storage

When a file is saved to a classical server, it resides whole. If the administrative boundary is breached, the cleartext contents are entirely compromised.

PERIMETER EXFILTRATION

Attackers targeting network interfaces to capture whole files in transit.

MULTI-TENANT
CONTAMINATION

Hardware-level bleeds inside shared hypervisors, exposing one customer's data to another.

CUSTODY KEY FORGERY

Forced access to a central database holding system-managed master encryption keys.

"Inaya" is derived from the Arabic term for absolute protection, vigilant care, and functional grace — the operating philosophy behind treating storage as an active guardian process rather than a passive database write.

SECTION 03

Client-Side Encryption & Sharding

Files never leave the user's device in a unified, readable form. Every step below happens locally — verified directly against the custody-sdk source, not assumed.

KEY DERIVATION

```
Kv = PBKDF2(passkey, salt, iterations = 100,000, HMAC-SHA256, len = 256)
```

The master passkey (P) and a fresh, cryptographically random 16-byte salt (S) derive a 256-bit AES key locally — the passkey itself is never transmitted anywhere, at any point, in either direction.

The file is then encrypted with AES-GCM-256 and the resulting ciphertext (C) is bisected at its exact midpoint into two shards:

BINARY SHARDING

```
Mp = floor(length(C) / 2)
Shard Alpha = C[0 ... Mp-1]
Shard Beta  = C[Mp ... end]
```

Neither shard alone contains a contiguous, decryptable structure — an isolated compromise of a single storage node yields random binary noise, not a usable file fragment. This is literal midpoint bisection, not erasure coding.

SECTION 04

On-Chain Ledger Attestation

Inaya eliminates the need for a central database cluster for asset ownership — the registry lives on-chain, tracking only shard pointers (IPFS CIDs) and a file hash, never file content.

REGISTRY WRITE (INAYACUSTODY)

```
function batchRegisterAssets(  
  bytes32[] fileHashes,  
  uint256[] fileSizes,  
  string[] shardACIDs,  
  string[] shardBCIDs  
) external
```

This write is permanent by design — there is no update or delete function for a registered asset. Renaming, moving, or deleting a file is an off-chain metadata operation, not a contract mutation.

SECTION 05

Pay-As-You-Go Storage

- Storage is billed in stablecoin (USDT) to remove multi-token friction for developers and retail users.
- Fee rates are read live from the deployed contract (`usdtFeePerGB` / `inayaFeePerGB`) rather than hardcoded — the protocol can adjust pricing over time without a client update.
- [VERIFY] Baseline rate and egress/retrieval pricing — prior published figures (4.5 USDT/TB/month storage, 5–10 INAYA per 0.5–1 TB egress) should be re-confirmed against the live contract values before being restated publicly; rates are designed to move.
- Staking \$INAYA unlocks priority bandwidth routing and a share of network fee yield (Section 09).

SECTION 06

Corporate Reserve — Annual Plans

For institutions with fixed large-scale storage needs. This is the real, implemented tier structure — three fixed annual allocations, not an API-call-volume tier system.

ALLOCATION	RESERVE FEE (USDT/YR)	ANNUAL MAINTENANCE (INAYA-EQ/YR)
250 TB / Year	13,500	500
500 TB / Year	27,000	1,000
1000 TB / Year	54,000	2,000

[VERIFY] These figures match the live pricing shown in the dApp's Business Model tab as of this writing — reconfirm before restating, since the UI reads them from config that can change.

SECTION 07

Revenue Distribution — RevenueRouter & Corporate Escrow

Corrected from the prior edition of this document, against the real transaction flow in the dApp's checkout code.

1. **1.** A corporate customer approves USDT and calls `RevenueRouter.processCorporateInvoice(usdtAmount)` — this is a real, standalone on-chain transaction.
2. **2.** The client then separately approves USDT and calls `InayaCorporateEscrow.createEscrow(corporate, node, cogsAmount)` for the node-operator's COGS share — a second, independent transaction, not an automatic cascade triggered by the router itself.
3. **3.** `InayaCorporateEscrow` locks that allocation and releases it over 12 fixed monthly payouts (`releaseMonthlyPayout()`, callable by anyone once a release is due).

Correction, stated plainly. An earlier version of this material described the router→escrow handoff as happening "atomically within the same blockchain transaction" with `RevenueRouter` "automatically" calling `createEscrow()`. That is not what the deployed contracts do — it's two separate transactions, both client-initiated. The revenue-split percentages themselves ([VERIFY] 39% node / 51% treasury / 10% team, if still current) are a business decision this document doesn't independently confirm from `RevenueRouter`'s source, since that contract's Solidity isn't tracked in this repository — only its deployed address and interface are known.

SECTION 08

Node Performance & Proof of Storage

- InayaNodeRegistry tracks each node's capacity, tier, uptime, and stake — but per the contract's own header comment, this is coordinator-verified telemetry (an authorized verifier wallet attests it), not cryptographic proof-of-storage. Described accurately here, not oversold.
- InayaProofRegistry separately stores a Merkle root per asset and verifies chunk-level proofs — today this verification is backend-checked (onlyOwner); the contract's own comment documents a planned path to make it permissionless and stake-slashing-backed, not yet built.
- Settlement is deliberately timelocked: a verifier queues a settlement (computing commission, not moving funds), and only after a 36-hour delay can anyone call releaseSettlement — protecting the reserve from a single compromised key.

SECTION 09

Token Economics

ALLOCATION	%	TOKENS
Swarm Reserve (Node Rewards)	40.0%	12,000,000 INAYA
Staking Rewards Pool	26.7%	8,000,000 INAYA
Liquidity Pool	21.7%	6,500,000 INAYA
Team Runway	5.0%	1,500,000 INAYA
Core Ecosystem Fund	3.3%	1,000,000 INAYA
Genesis Airdrop	3.3%	1,000,000 INAYA

[VERIFY] Fixed 30,000,000 total supply and the allocation split above are stated as previously published — this document doesn't independently re-derive them from on-chain state, since \$INAYA's Solidity source isn't tracked in this repo (only its deployed address). Confirm before external distribution.

Swarm Reserve emissions are uptime-gated, not automatic:

- [VERIFY] A 3-month, 90%-uptime commitment cliff before a node qualifies for any Swarm Reserve emission.
- [VERIFY] Monthly cap by tier: 30 \$INAYA (98%+ uptime), 20 \$INAYA (95–97.9%), 10 \$INAYA (90–94.9%), 0 below 90%.
- These specific thresholds are a tokenomics design decision, not something the current contracts enforce directly (InayaNodeRegistry's tier/commission logic is confirmed in code; the emission-cap schedule above should be checked against whatever governs actual \$INAYA distribution before being restated as fact).

SECTION 10

Beyond The Protocol — The Application Layer

Everything above is the foundation. Since the protocol's original whitepaper, a full application layer has shipped on top of it — most of which a user never has to know is running on a blockchain at all.

- Business Workspace — a B2B SaaS product: organizations, departments, projects, and documents with server-enforced approval workflows, granular permissions, secure sharing, and a permission-aware AI assistant. Email sign-in, no wallet required.
- Security Layer ("Inaya Firewall") — a decentralized, node-reported threat-intelligence network with reputation-weighted confirmation and on-chain-anchored verdicts, surfaced on a public web page, in the mobile app, and via real OS-level enforcement on desktop.
- Inaya Learn — a YouTube-based educational discovery platform with an AI tutor, built to make the apps a daily-use destination beyond storage and staking.
- Investor Data Room — a branded, access-controlled document room with per-visitor engagement tracking.
- Two Tauri desktop apps — thin native wrappers around the Business Workspace and the main dApp, with system tray, native notifications, and signed auto-updates.
- Three Gemini-powered AI assistants — Business, Security, and Learn — sharing one technical pattern but opposite guardrail philosophies suited to their purpose (see the Ecosystem Architecture document for full detail).

Full technical depth on every one of these lives in the companion "Complete Ecosystem Architecture" document — this whitepaper stays focused on the core protocol.

SECTION 11

Leadership

Talha Waqas — Founder & CTO

Web3 full-stack engineer specializing in client-side cryptographic infrastructure, EVM smart contract architecture, and secure data-routing pipelines. Drives core protocol development end to end.

Yakub Adnan — Co-Founder & Growth Lead

Web3 growth operator and community strategist specializing in DePIN, user acquisition, and AI-driven ecosystem scaling. Leads growth architecture, community operations, and campaign distribution.

Fibha Urooj — CFO

Leads financial planning, budgeting, compliance, and operational finance — building the financial foundation supporting Inaya Network's long-term growth.

SECTION 12

Conclusion

Absolute Protection, Vigilant Care, and Functional Grace — the guiding principles of a network built to guard what matters most.

The Inaya Protocol demonstrates that data integrity doesn't require a centralized database — it can be achieved through client-side encryption, decentralized storage, and a public, tamper-evident ledger. Everything described in this document runs today on BNB Chain Testnet.